

Project 1 - Part 1

Student Information Management System (SIMS)

Programmer: Branson Zemaitis

Testing Requirements

All three required test cases (Alice Johnson, Michael Brown, Christopher Williams) were run against the compiled program and produced output matching the PDF's expected output exactly. Screenshots of each run follow on the next pages.

Bugs Encountered and How They Were Corrected

Bug: When run through certain IDEs (e.g. VS Code's built-in code runner) instead of a real terminal, the three input prompts ("Enter Student ID:", "Enter Student Name:", "Enter Current GPA:") all appeared jumbled together on screen before any input could be typed, instead of appearing one at a time in order.

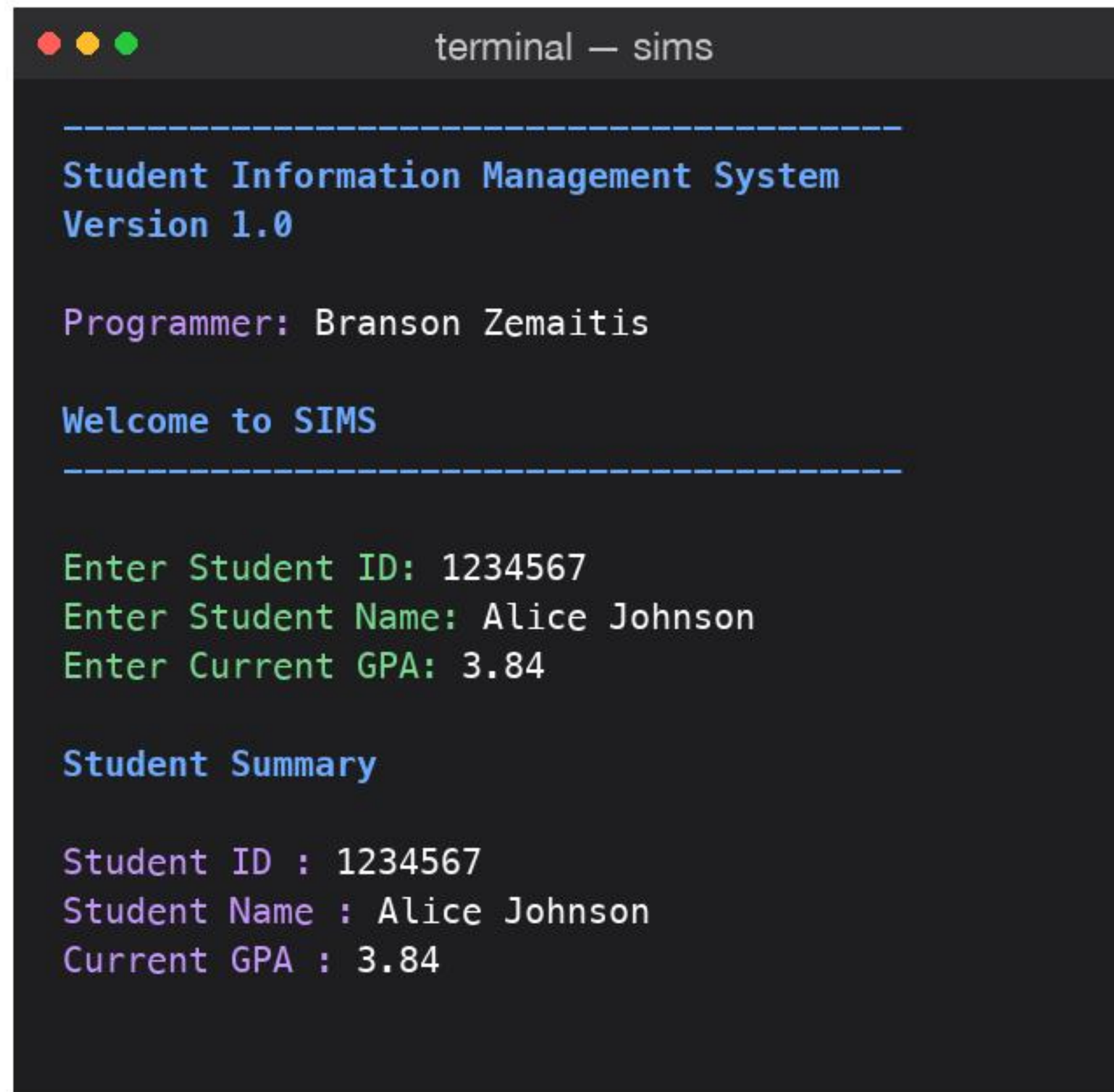
Cause: C's standard output stream is only "line-buffered" (printed immediately after each newline) when connected to a real terminal. When it is instead connected to a non-terminal output (such as some IDE run panels), it switches to "fully buffered" mode, holding printed text in memory until the buffer fills or the program ends, instead of showing it right away. Since each prompt was printed without a trailing newline (so the user could type on the same line), the prompt text stayed stuck in the buffer instead of appearing before the program paused to wait for input.

Fix: Added a call to `fflush(stdout);` immediately after each prompt's `printf()` statement. This forces any buffered output to print to the screen right away, regardless of how the program is being run, so each prompt always appears before the program waits for input.

Verification: Recompiled with `-Wall -Wextra` (no warnings) and re-ran all three required test cases; output still matched the expected output exactly after the fix.

Test Case 1 — Alice Johnson (ID 1234567, GPA 3.84)

Result: PASS — output matches PDF's expected output exactly.



```
terminal — sims

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Student Information Management System
Version 1.0

Programmer: Branson Zemaitis

Welcome to SIMS
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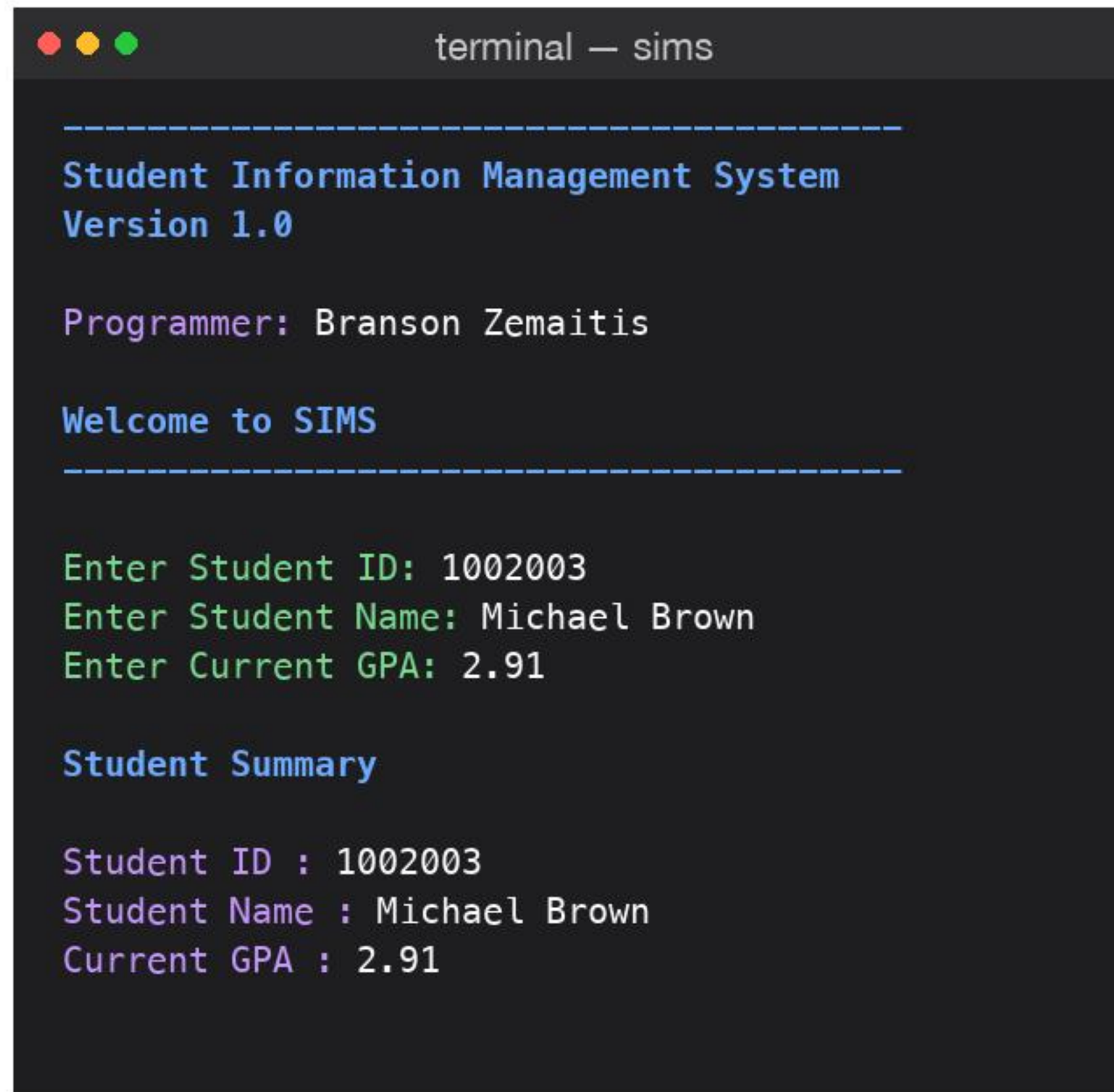
Enter Student ID: 1234567
Enter Student Name: Alice Johnson
Enter Current GPA: 3.84

Student Summary

Student ID : 1234567
Student Name : Alice Johnson
Current GPA : 3.84
```

Test Case 2 — Michael Brown (ID 1002003, GPA 2.91)

Result: PASS — output matches PDF's expected output exactly.



```
terminal — sims

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Welcome to SIMS
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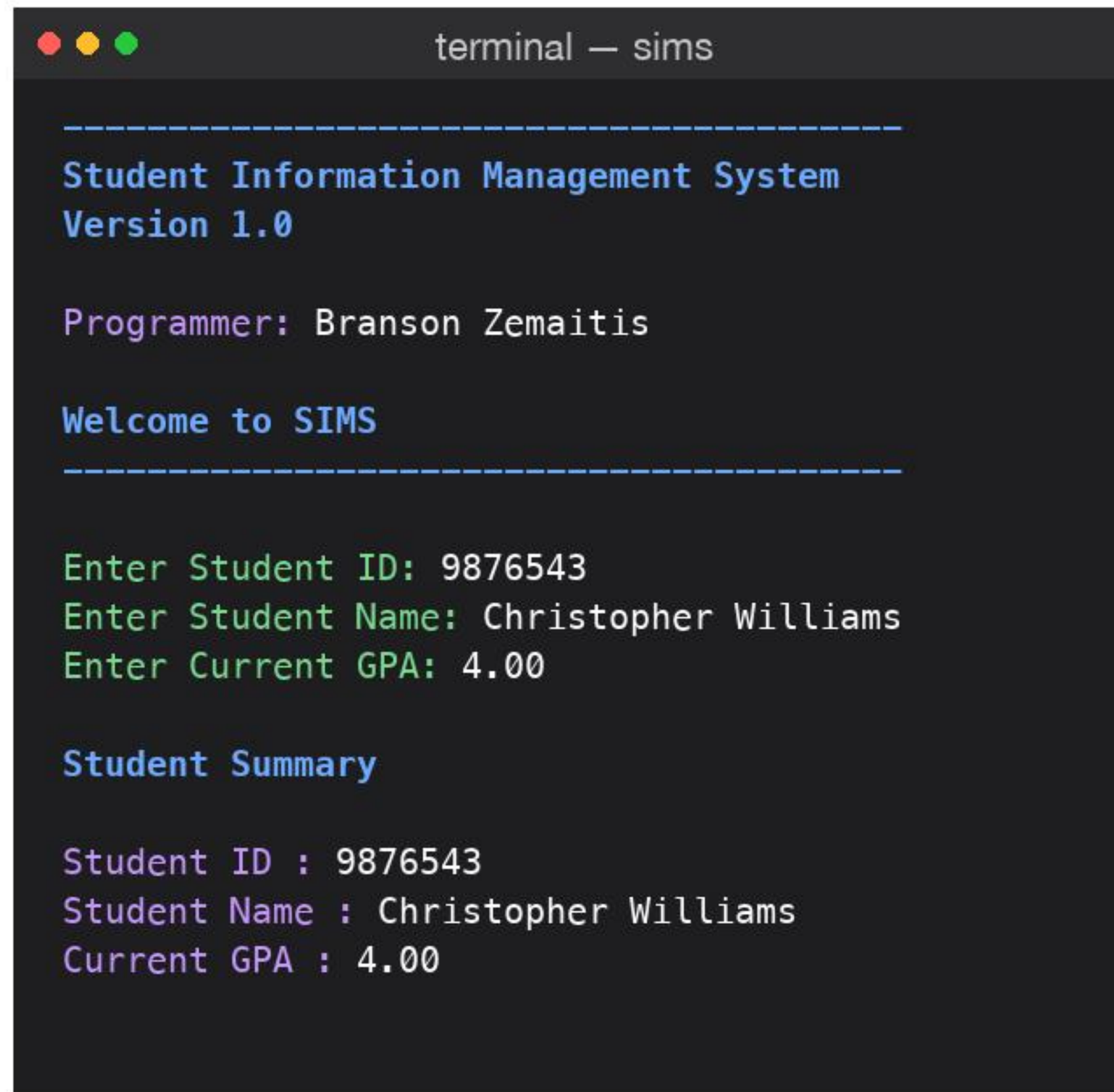
Enter Student ID: 1002003
Enter Student Name: Michael Brown
Enter Current GPA: 2.91

Student Summary

Student ID : 1002003
Student Name : Michael Brown
Current GPA : 2.91
```

Test Case 3 — Christopher Williams (ID 9876543, GPA 4.00)

Result: PASS — output matches PDF's expected output exactly.



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terminal — sims

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Welcome to SIMS
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Enter Student ID: 9876543
Enter Student Name: Christopher Williams
Enter Current GPA: 4.00

Student Summary

Student ID : 9876543
Student Name : Christopher Williams
Current GPA : 4.00
```